

Arthur Chien

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EDUCATION

Carnegie Mellon University, School of Computer Science Pittsburgh, PA
Master of Science in Artificial Intelligence and Innovation (MSAI), GPA: 4.0/4.0 August 2024 – May 2026

National Chengchi University Taipei, Taiwan
*Bachelor of Science in Management Information Systems, GPA: 4.0/4.3, Dean's List * 4* August 2019 – January 2024

Coursework: Operating Systems, Data Structures, Algorithms, Computer Networks, Databases, Distributed Systems, Statistics, Discrete Mathematics, Linear Algebra, Machine Learning, Deep Learning, Natural Language Processing

SKILLS

Programming Languages: Python, Java, C/C++, SQL, JavaScript (React, Node.js), HTML/CSS, R
AI & Agents: LLMs, OpenHands, LangGraph, LangChain, RAG, Multi-modal, Evaluation
Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Transformers, CNNs, GNNs, MLflow
Backend & DevOps: Linux, Docker, Kubernetes, Apache Kafka, RESTful APIs, PostgreSQL, MongoDB, Redis, AWS, GCP, Azure, Git/GitHub, CI/CD (Actions, Jenkins), Prometheus, Grafana, OpenTelemetry, Elasticsearch, Kibana

WORKING EXPERIENCE

AI Engineer & Capstone Researcher Pittsburgh, PA, USA
Bank of New York Mellon (BNY) & Carnegie Mellon University September 2025 – May 2026

- Built a **digital employee platform** with **React** and **FastAPI** that compiles user-defined roles, tools, and skills into runnable **OpenHands** agents, reducing manual build time from **3 months to 3 minutes**
- Engineered a **multi-modal skill ingestion pipeline** that extracts actionable tool schemas and workflows from **video and audio demonstrations**, enabling agents to learn reusable skills from human examples
- Developed an agent reasoning loop with **self-reflection** and trajectory replay, boosting task success rate by **21%**
- Built an evaluation framework using **Docker sandboxes** to benchmark agents on task/step success and latency

Software Engineer Intern Hsinchu, Taiwan
Taiwan Semiconductor Manufacturing Company (TSMC) June 2025 – August 2025

- Developed automated test case generation from **OpenTelemetry** traces by constructing execution-flow graphs and analyzing call paths, achieving **88%** recall of failing services in pre-production
- Parsed flowchart XMLs into graphs to generate natural language docs with LLMs for a **RAG-based** AI assistant

Research Assistant Taipei, Taiwan
Data Mining and Machine Learning Laboratory, Academia Sinica April 2023 – July 2023

- Built a document similarity module using **SimGNN** with **GCNs** and **SimHash**, improving scientific paper recommendations by **30%** and boosting link prediction accuracy by **12%** on a 1M+ document dataset

PROJECTS

Scalable Movie Recommendation System with Monitoring and Continuous Deployment

- Built a scalable movie recommendation system for **1M+ users** with **Kafka** for real-time ingestion, **MongoDB** for storage, and containerized **RESTful API** microservices using **Docker** and **Kubernetes**
- Automated model retraining and deployment every **48h** via **Jenkins** pipelines and versioned models in **MLflow**
- Monitored system metrics via **Prometheus** and **Grafana**, including click-through rate (CTR), latency, and alerts

Latent Guard++: Context-Aware Safety Framework for Generative Models

- Built a hybrid safety filtering system for **20k+ text-to-image prompts**, enhancing baseline Latent Guard with LLM-guided context awareness and tested across in-/out-of-distribution and adversarial datasets
- Proposed **dynamic thresholding** that adapts classification boundaries using **LLM-estimated risk confidence**
- Achieved **13.9%** accuracy improvement on **unseen datasets** and **10% gain** on adversarial prompts

Automated Hit-frame Detection for Badminton Match Analysis

- Implemented a **microservices**-based badminton analysis system using **PyTorch**, **OpenCV**, and **Flask**, processing **500+** professional match videos and reducing manual analysis time by **75%**, as presented in a [first-author paper](#)
- Architected a **CNN**-based rally segmentation and player tracking pipeline, achieving **89% precision** in rally identification and **97.5% PCK@0.2** for in-court player detection with a custom-trained **Keypoint-RCNN**
- Developed a seq2seq **Transformer**-based shuttle direction prediction model, achieving **96% accuracy**